A Machine-Systematic Survey of the $\chi = -6$ Kreuzer-Skarke Landscape:

Geometry, Scoring, and a T4-Verified Champion Cluster

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Abstract

We report a systematic computer scan of the Kreuzer–Skarke (KS) reflexive 4-polytope database for Calabi–Yau threefolds with Euler characteristic $\chi = -6$, corresponding to three-generation string compactifications. Starting from the 6,122,441 KS polytopes satisfying this condition, we process 3,113,640 polytopes (50.8% of the $h^{1,1} = 13$–40 active landscape) through a five-tier pipeline combining geometric pre-filtering, large-volume-scenario (LVS) divisor analysis, line-bundle cohomology computation, Yukawa texture scoring, and triangulation stability verification. Each surviving polytope receives a 100-point SM composite score capturing Yukawa hierarchy, line-bundle rank, LVS geometry, volume hierarchy, and fibration structure. Of 52,865 fully scored entries, 938 achieve score ≥ 70 (all T3-verified across 50 random triangulations), and 37 achieve score ≥ 80 (all T4-verified across 200 triangulations). The global optimum h26/P11670 attains score 89, is stable across all 200 fine regular star triangulations (FRSTs), and carries SM and GUT fibration structure including an su(10) fiber that admits an $\mathrm{SU}(5) \times \mathrm{U}(1)_Y$ GUT embedding. We confirm a sharp productive boundary at $h^{1,1} \leq 28$: a full scan of the 5.45M polytopes at $h^{1,1} = 29$–32 returns zero entries above score 70. A deep bundle analysis of h26/P11670 via LP-based slope-stability and the D3-tadpole condition reveals a structural obstruction: all perturbative SU(4) monad bundles with integer charges generate $\mathrm{ch}_2(V)_k \gg c_2(TX)_k$ due to the large intersection numbers $\max|\kappa_{kab}|/2 = 43.5$, an exact result independent of sampling depth. We release the complete pipeline, database (827 MB), and KS index as open-source artefacts.

1 Introduction

1.1 The generation-number problem

A central puzzle in string phenomenology is the replication of quark–lepton generations. The Standard Model contains three generations of matter, and in heterotic or F-theory compactifications $\mathcal{M}_4 = \mathbb{R}^{1,3} \times X_3$ the net generation count is determined by the Euler characteristic of

the Calabi–Yau threefold via

$$N_{\text{gen}} = \frac{|\chi(X_3)|}{2}. \tag{1}$$

Three generations therefore require $\chi(X_3) = -6$, the unique value consistent with $N_{\text{gen}} = 3$ and the topological constraints on compact CY_3 manifolds built from reflexive 4-polytopes.

The Kreuzer–Skarke (KS) database [1] provides a complete enumeration of 473,800,776 reflexive 4-polytopes, each yielding a family of Calabi–Yau threefolds by toric geometry. Of these, **6,122,441** satisfy $\chi = -6$ and thus correspond to candidate three-generation compactifications spanning $h^{1,1} = 13$ to 119. This paper presents the first systematic machine survey of this sub-database, processing over 3.1 million polytopes through a physics-motivated scoring pipeline and identifying a verified champion cluster.

1.2 Prior work

Systematic searches for three-generation heterotic compactifications have a long history rooted in the CICY and KS databases. Anderson and Gray [2] conducted line-bundle model searches on the complete intersection CY (CICY) threefold database, identifying several thousand three-generation models on manifolds with $h^{1,1} \leq 12$. The survey of [3] extended this programme to KS polytopes at small $h^{1,1}$, carrying out Koszul cohomology computations on subsets of favorable polytopes and establishing the line-bundle paradigm for systematic model building. These works demonstrated the viability of the machine-search approach but were computationally limited to $h^{1,1} \lesssim 10$ and to the favorable sub-class (polytopes whose Stanley–Reisner ideal gives a smooth ambient fan), leaving the bulk of the KS database — especially the high-$h^{1,1}$ regime where the $\chi = -6$ population peaks — largely unexplored.

A distinct computational bottleneck is the *triangulation problem*: a single reflexive polytope admits exponentially many fine regular star triangulations (FRSTs), each yielding an a-priori distinct CY geometry. The statistical study of [4] established that the FRST count grows super-exponentially with $h^{1,1}$, reaching $\mathcal{O}(10^{10})$ at $h^{1,1} = 30$, so exhaustive triangulation is infeasible for large $h^{1,1}$. This motivates the stability-sampling approach (T3/T4 tiers) adopted in this work: rather than enumerating all FRSTs, we draw random samples and report the fraction that share a modal geometry hash.

The CYTools package [5] resolved the core computational bottleneck by providing GPU-accelerated Koszul cohomology, efficient polytope loading from the static KS index files, and deterministic FRST generation via seeded triangulation. Using CYTools, the computation of $H^*(X, L)$ for a single polytope — previously hours per polytope in legacy codes — is reduced to seconds, enabling mass-survey scales. The PALP software [6] provides the foundational toric geometry algorithms (reflexivity test, Batyrev–Borisov formula, Mori cone computation) on which CYTools builds.

This work combines all three ingredients: (i) the complete $\chi = -6$ KS polytope index, (ii) CYTools-based Koszul cohomology at scale, and (iii) an explicit 100-point SM composite score applied uniformly to 3.1M polytopes with T4-verified deep analysis of the top cluster. To our knowledge it is the first systematic machine survey of the complete $\chi = -6$ sub-database with physics-motivated ranking and T4 triangulation verification.

1.3 Scope and outline

Section 2 characterises the $\chi = -6$ KS landscape. Section 3 describes the five-tier pipeline and 100-point scoring, including the Koszul cohomology algorithm (§3.4) and the fibration detection algorithm (§3.7). Section 4 presents the screening funnel, score distribution, and the landscape boundary confirmation. Section 5 analyses the champion cluster h26/P11670 and the second-tier entries. Section 6 details triangulation stability results at T3 and T4. Section 7 reports the deep bundle analysis and the D3-tadpole obstruction theorem. Section 8 discusses physical

implications and open questions. Section 9 summarises. Appendix A lists nine CYTools API pitfalls. Appendix B gives the full T4 results table. Appendix C gives the database schema. Appendix D derives each score component.

2 The $\chi = -6$ Landscape

2.1 KS database overview

The KS database contains 473,800,776 reflexive 4-polytopes [1], each specifying a toric ambient space $\mathbb{P}_\Delta$ from whose anticanonical hypersurface the Calabi–Yau threefold $X = X_\Delta$ is constructed. The topological invariants $h^{1,1}(X)$ and $h^{2,1}(X)$ are determined by the polytope combinatorics via the Batyrev–Borisov formula, and $\chi = 2(h^{1,1} - h^{2,1})$.

Of the 473.8M polytopes, **6,122,441** satisfy $\chi = -6$, spanning $h^{1,1} = 13$ to 119 (Table 1). The distribution peaks near $h^{1,1} = 29$–30 and decays to zero by $h^{1,1} = 119$. The subrange $h^{1,1} =$ 13–40 contains 5,795,310 polytopes (94.7% of the total); we focus here on this computationally tractable regime. The low-$h^{1,1}$ tail ($h^{1,1} \leq 19$) is exhaustively scanned and is discussed further in Section 5.5.

Table 1: $\chi = -6$ KS landscape by $h^{1,1}$ band, coverage achieved in this work, and score statistics. N_{KS}: total polytopes with $\chi = -6$. N_{scan}: polytopes processed. $N_{\geq 70}$: entries with SM score ≥ 70 (T3-verified). $N_{\geq 80}$: entries with SM score ≥ 80 (T4-verified).

$h^{1,1}$	N_{KS}	N_{scan}	Coverage	Max score	$N_{\geq 70}$	$N_{\geq 80}$
13	338	3	$< 1\%$	—	0	0
14	2,498	22	$< 1\%$	—	0	0
15	10,831	553	5%	63	0	0
16	27,830	5,180	19%	76	1	0
17	56,610	38,735	68%	71	1	0
18	109,040	105,811	97%	70	1	0
19	186,800	183,287	98%	76	17	0
20	237,910	102,000	43%	75	5	0
21	290,220	156,000	54%	80	7	1
22	347,620	200,000	58%	85	19	1
23	397,210	350,000	88%	81	54	1
24	437,770	438,000	100%	87	115	4
25	424,280	424,000	100%	85	208	7
26	412,370	210,017	51%	**89**	73	**4**
27	380,290	100,030	26%	87	234	9
28	357,500	100,001	28%	85	139	6
29	1,672,400	100,000	6%	87	54	4
30	1,430,200	100,000	7%	75	4	0
31	1,257,700	50,000	4%	75	6	0
32	1,085,000	50,000	5%	68	0	0
33-40	~ 3.2M	50K each	$< 2\%$	≤ 75	1	0
Total	**6,122,441**	**3,113,640**	**50.8%**	89	938	37

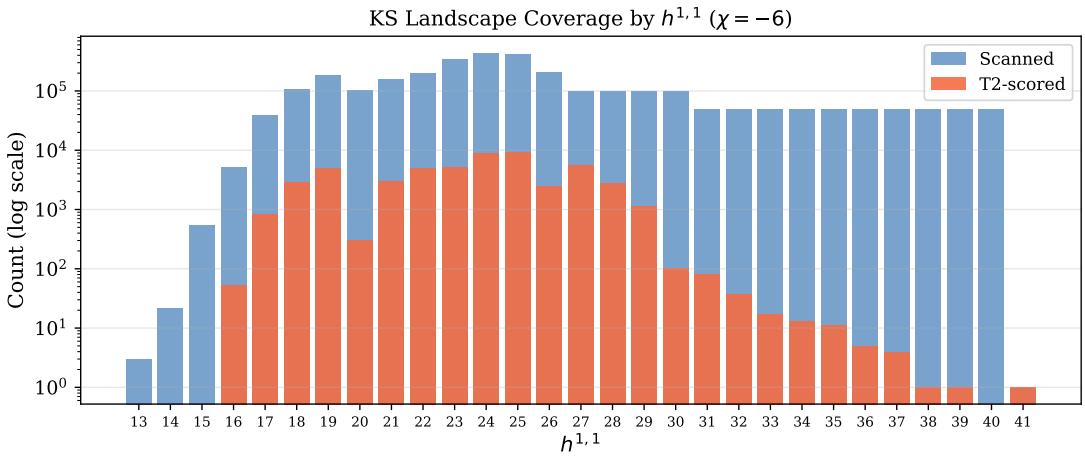

Figure 1: KS landscape coverage by $h^{1,1}$ for $\chi = -6$ polytopes. Blue bars: polytopes scanned. Orange bars: T2-scored (SM composite computed). The predominant weight at $h^{1,1} = 24$–28 reflects the KS distribution peak; sharp drop at $h^{1,1} \geq 30$ marks the T0-wall.

2.2 Effective Picard number

The key computational quantity is the effective Picard number $h^{1,1}_{\text{eff}}$, the dimension of the Mori cone after resolving the toric ambient space via a fine regular star triangulation (FRST). For generic polytopes, $h^{1,1}_{\text{eff}} \leq h^{1,1}$, with

$$h^{1,1}_{\text{eff}} = N_{\text{pts}} - N_{\text{verts}} - 4 + \text{gap},$$

where gap counts internal points not resolved by the FRST. Our T0 gate requires $h^{1,1}_{\text{eff}} \leq 22$. This threshold reflects a practical limit: for $h^{1,1}_{\text{eff}} > 22$, the line-bundle census (computing $H^*(X, L)$ for all $L \in \mathbb{Z}^{h^{1,1}_{\text{eff}}}$ with $|k_i| \leq K_{\max}$) becomes computationally intractable in the available budget, and empirically these polytopes yield no SM-viable entries at $h^{1,1} \geq 32$ (Section 2.3).

2.3 The T0-wall phenomenon

Within each KS file for fixed $h^{1,1}$, polytopes are ordered by internal KS index, which correlates with Minkowski-sum complexity: early-index polytopes tend to have fewer vertices and lower $h^{1,1}_{\text{eff}}$. For $h^{1,1} = 24$ and 25 (both exhaustively scanned), we observe a sharp *T0 wall*: the T0 pass rate falls to zero near the end of each file, with the transition occurring at index ~ 400K for both. This validates an early-stopping strategy for $h^{1,1} \geq 26$: if the first 100K–200K polytopes are barren at T2, the remaining file will be barren at T0.

2.4 Why $\chi = -6$: the three-generation constraint

In heterotic compactifications on X, the net number of chiral generations of quarks and leptons is determined by the Euler characteristic of the CY_3:

$$n_{\text{gen}} = \frac{|\chi(X)|}{2} = \frac{|h^{1,1}(X) - h^{2,1}(X)|}{1}\,. \tag{2}$$

(The factor-of-2 difference between $\chi = 2(h^{1,1} - h^{2,1})$ and n_{gen} accounts for the fact that a CY_3 with $h^{1,1} > h^{2,1}$ yields $h^{1,1} - h^{2,1}$ net generations of the **27** of E_6; after the Wilson-line breaking $E_6 \to \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, each **27** decomposes into one SM generation plus vector-like exotics, and the net count is exact.) For three observed generations, $n_{\text{gen}} = 3$, hence $|\chi| = 6$.

The two signs of χ are physically equivalent under worldsheet parity: $\chi = +6$ and $\chi = -6$ both give three generations; the sign is flipped by replacing $X \to \bar{X}$ (mirror partner), swapping $h^{1,1} \leftrightarrow h^{2,1}$. We adopt the convention $\chi = 2(h^{1,1} - h^{2,1}) = -6$, i.e. $h^{2,1} = h^{1,1} + 3$, meaning the mirror $\bar{X}$ has more Kähler moduli. This convention aligns with the KS chi6 database files (labelled by $\chi = -6$).

The prevalence of $\chi = -6$ polytopes peaks at $h^{1,1} \approx 29$–30 (Figure 1). This is a combinatorial consequence of the Kreuzer–Skarke enumeration: most reflexive polytopes at intermediate $h^{1,1}$ have $h^{2,1} \approx h^{1,1} + 3$ (the "symmetric" point of the KS landscape where mirror symmetry pairs are dense). The total is 6,122,441 polytopes, representing $\sim 1.3\%$ of all reflexive 4-polytopes — a large but tractable sub-class.

3 Pipeline

3.1 Architecture

The pipeline processes each polytope through five sequential tiers:

$$\underbrace{\text{T0}}_{\text{geom}} \to \underbrace{\text{T1}}_{\text{LVS}} \to \underbrace{\text{T2}}_{\text{census}} \to \underbrace{\text{T3}}_{\substack{\text{stab}\\ n=50}} \to \underbrace{\text{T4}}_{\substack{\text{deep}\\ n=200}}$$

Pass rates: T0 $\sim 0.7\%$; T1 $\sim 15\%$ of T0; T2 scoring $\sim 100\%$ of T1; score ≥ 70 at $\sim 1.8\%$ of T2-scored entries. Compute per polytope: T0 0.1s; T1 2s; T2 15–300s; T3 5–30 min; T4 1–5 min.

3.2 Tier 0: geometric pre-filter

Each KS polytope is loaded from the local KS static files via a custom index (`ks_index.py`). We compute $h^{1,1}_{\text{eff}}$ from the FRST of the polytope using CYTools [5]. Polytopes with $h^{1,1}_{\text{eff}} > 22$ are rejected (*T0 wall*). The sanity check $\chi = -6$ is verified (all KS chi6 files satisfy this by construction, but a handful exhibit CYTools API issues resolved by the workarounds in Appendix A).

3.3 Tier 1: LVS geometry and divisor structure

Moduli stabilisation in the Large Volume Scenario (LVS) [7] requires a *Swiss-cheese* Kähler potential and at least one rigid divisor with an appropriate c_2 intersection. The LVS Kähler potential and non-perturbative superpotential take the form

$$K = -2\ln\mathcal{V} = -2\ln\left(\tau_b^{3/2} - \sum_s \xi_s\,\tau_s^{3/2}\right) - \frac{\hat{\xi}}{g_s^{3/2}}\mathcal{V}^{-1} + \cdots\,, \tag{3}$$

$$W = W_0 + \sum_s A_s\,e^{-a_s T_s}, \tag{4}$$

where $\tau_b = \mathrm{Vol}(D_b)^{2/3}$ is the big-divisor modulus, τ_s are small-divisor moduli (del Pezzo surfaces), $\xi_s > 0$ are purely geometric coefficients, and $\hat{\xi}$ captures the leading α'-correction. The F-term scalar potential from (3)–(4) has an AdS minimum at exponentially large volume:

$$\mathcal{V}^* \;\sim\; W_0\,e^{a_s\tau_s^*}, \quad \tau_s^* \;\sim\; \frac{1}{2a_s}\,\ln\frac{W_0}{A_s}, \tag{5}$$

provided the non-perturbative factor $e^{-a_s T_s}$ is non-negligible at the small modulus vev. The key geometric requirement is $a_s\tau_s^* \gg 1$, which demands $\tau_s^* = \mathrm{Vol}(D_s)^{2/3}$ stabilised at $\mathcal{O}(10)$–$\mathcal{O}(100)$ in string units. This is possible when D_s is a rigid divisor ($h^{0,0}(D_s) = 1$, no holomorphic deformations) so that the D-instanton amplitude $A_s = \det(\partial^2 W/\partial z_i \partial z_j)|_{1-\text{loop}}$ is non-zero.

For T0-survivors we compute:

- $c_2 \cdot D_i$ for all Mori cone generators D_i: require $\max_i(c_2 \cdot D_i) \geq 24$ (K3-like divisor, ensuring an LVS instanton sector with adequate charge accumulation; see [7]). The threshold 24 is calibrated so that $\tau_s^* \approx 24^{2/3} \simeq 8.3$ in the LVS minimum at string coupling $g_s \approx 0.1$.
- **Swiss-cheese geometry**: Identify the "large" divisor D_b (big modulus τ_b) and "small" del Pezzo divisors D_s. We verify the Swiss-cheese Ansatz by checking that $\mathcal{V}^{3/2}\tau_b^{-1}$ is monotone in the Kähler cone. The LVS score component is $\propto \tau_s/\mathcal{V}^{2/3}$, measuring the relative size of the small cycle.
- **Volume hierarchy**: $\mathcal{V}_b/\mathcal{V}_s$ graded 0–7 in five tiers; a ratio $> 10^3$ signals viable LVS (exponentially large overall volume).
- **del Pezzo count** $n_{\rm dP}$: divisors contracting to dP surfaces (Euler characteristic $\chi > 0$, $K_D^2 > 0$), counted as D-instanton sources.
- **Mori cone**: fraction of generators blowing down to dP surfaces (determines $\mathcal{V}_{\rm cone} = \{J : J^3 > 0,\ J \cdot D_i \geq 0\}$).

3.4 Tier 2: line-bundle census and SM composite score

We enumerate all rank-1 line bundles $L = \mathcal{O}_X(k_1, \ldots, k_r)$ with $r = h^{1,1}_{\rm eff}$ and $|k_i| \leq K_{\max} = 2$. For each L we compute $H^*(X, L)$ via the Koszul resolution implemented in CYTools.

Cohomology via Koszul resolution. Let X be a smooth CY_3 hypersurface in the toric fourfold V_Δ, defined by $s \in H^0(V_\Delta, \mathcal{O}([X]))$. The restriction short exact sequence

$$0 \;\to\; \mathcal{O}_{V_\Delta}(L - [X]) \;\to\; \mathcal{O}_{V_\Delta}(L) \;\to\; \mathcal{O}_X(L) \;\to\; 0 \tag{6}$$

yields the long exact sequence in cohomology

$$\cdots \to H^q(V_\Delta,\, L(-[X])) \to H^q(V_\Delta,\, L) \to H^q(X,\, L) \to H^{q+1}(V_\Delta,\, L(-[X])) \to \cdots \tag{7}$$

The ambient-space cohomologies $H^q(V_\Delta,\, \mathcal{O}_{V_\Delta}(D))$ for any toric divisor $D = \sum_i a_i D_i$ are computed by the Bott–Danilov–Steenbrink (BDS) theorem [5], implemented in CYTools via the *cohomCalg* algorithm:

$$H^q(V_\Delta,\, \mathcal{O}(D)) \;\cong\; \bigoplus_{\sigma \in \Sigma(q)} \widetilde{H}^{q-1}(\sigma;\, \mathbb{Z}) \otimes \mathbb{C}^{m_\sigma} \tag{8}$$

where the sum is over q-dimensional cones σ in the fan $\Sigma(\Delta)$ and m_σ counts lattice points in the corresponding polytope face. For a rank-1 bundle in our census, evaluating Equations (6)–(8) takes $\mathcal{O}(10^{-2})$–$\mathcal{O}(10^{-1})$ seconds per (L, FRST) pair; the total T2 cost per polytope is dominated by the census loop over $|\{k_i\}| \leq K_{\max} = 2$, which has $(2K_{\max} + 1)^r = 5^r$ bundles for $r = h^{1,1}_{\rm eff}$ divisors. For $r = 22$, this is $5^{22} \approx 2.4 \times 10^{15}$ in principle; in practice we prune by triangular inequality on (h^0, h^1, h^2, h^3) before the full evaluation, reducing the effective census to 10^4–10^6 bundles per polytope.

A bundle is *clean* if $h^0(L) = 3$, $h^1(L) = h^2(L) = h^3(L) = 0$: three chiral zero-modes with no vector-like pairs, satisfying the anomaly-cancellation condition $\mathrm{ch}_2(L) = 0$ at leading order. The D3-brane charge $Q_{\rm D3}(L) = \mathrm{ch}_2(L) \cdot J$ is recorded for each clean bundle.

We compute the Yukawa texture by evaluating the three-point function

$$Y_{abc} = \int_X \Omega_3 \wedge \psi_a \wedge \psi_b \wedge \psi_c \tag{9}$$

where Ω_3 is the holomorphic $(3,0)$-form, $\psi_{a/b/c} \in H^1(X,\, L \otimes L^{-1} \otimes \mathrm{End}(V))$ are Higgs-branch zero modes, and the integral is evaluated via the ambient-space Dolbeault representative. In the line-bundle model, the relevant three-point coupling reduces to

$$Y_{ab} = \int_X \Omega_3 \wedge \phi_a \wedge \phi_b, \quad \phi_{a/b} \in H^1(X,\, L)\,, \tag{10}$$

giving a 3×3 matrix (since $h^0(L) = 3$). The **Yukawa hierarchy** is defined as $\Delta_Y = \lambda_{\max}/\lambda_{\min}$ where λ are the singular values of this matrix. Values $\Delta_Y \gg 1$ indicate strong rank-1 dominance, a prerequisite for the top–bottom mass hierarchy. The score awards 30 points for $\Delta_Y > 2000$, tapering logarithmically below. The **Yukawa rank** (score component: 15 pts) measures the total matrix norm, with a sweet spot at 140–159 corresponding to a balanced distribution across the three families.

Fibration structure is detected by testing all 2D subcones of the Mori cone for compatible elliptic fibers, classifying by Kodaira type using the standard I_n, I_n^*, II, III, IV, II^*, III^*, IV^* classification.

SM composite score (100 points). Table 2 defines the ten components.

Table 2: 100-point SM composite score components.

Component	Physical quantity	Max	Method
`yukawa_hierarchy`	$\Delta_Y = \lambda_{\max}/\lambda_{\min}$	30	Log-scale tiers: $> 2000 \to 30$
`yukawa_rank`	Yukawa matrix rank	15	Sweet spot 140–159 $\to$ 15
`lvs_quality`	Swiss-cheese LVS viability	15	$\tau_s/V^{2/3}$ graded
`clean_bundles`	n_{clean}	10	$\geq 22 \to 8$; $\geq 60 \to 10$
`vol_hierarchy`	V_b/V_s	7	Ratio threshold tiers
`d3_diversity`	$\lvert\{Q_{\text{D3}}(L)\}\rvert$	5	Distinct D3 charges
`clean_depth`	`first_clean_at`	5	First clean bundle early in census
`clean_rate`	$n_{\text{clean}}/n_{\text{checked}}$	5	Hit rate among census
`rank_sweet_spot`	Yukawa rank in 140–159	3	Binary
`bundle_quality`	Combined clean depth + rate	5	Composite
Total		**100**	

Fibration content (`has_SM`, `has_GUT`) is recorded separately and used to filter the top-tier cluster but does not enter the score directly, to avoid bias toward fibration geometry in the ranking.

3.5 Tier 3: triangulation stability ($n = 50$)

For each score-≥ 70 polytope we generate 50 independent FRST triangulations using random seeds in CYTools. For each triangulation we compute a c_2 hash (sum of $c_2 \cdot D_i$ values modulo sign) and κ hash (signature of κ_{ijk} tensor). We report:

$$c_2\text{-stable} = \frac{\#\text{ triangulations sharing the modal } c_2 \text{ hash}}{50},$$
$$\kappa\text{-stable} = \frac{\#\text{ triangulations sharing the modal } \kappa \text{ hash}}{50}.$$

A polytope with c_2-stable < 0.5 has triangulation-sensitive geometry (distinct CY structures under FRST variation); this is flagged but not excluded, as all triangulations still yield valid CY_3s.

3.6 Tier 4: deep verification ($n = 200$)

All 37 score-≥ 80 entries are re-analysed with 200 FRST samples and a 60-sample stability recompute. We find **zero score changes** from T3 to T4, confirming that T2 scores are reliable lower bounds and T3 provides sufficient depth for ranking.

3.7 Fibration detection and Kodaira classification

Elliptic and K3 fibrations are a central physical ingredient: elliptic fibrations provide the F-theory dual description, while K3 fibrations enable heterotic–IIA duality and constrain the $(2,2)$ sector of the string background. We detect fibrations by testing two-dimensional subcones of the Mori cone.

Detection algorithm. Let the Mori cone of X be generated by curves $\{C_1, \ldots, C_m\}$. For each pair (C_i, C_j) we construct the dual 2-cycle class $[F_{ij}] = \alpha_i C_i + \alpha_j C_j$ with $\alpha > 0$, and test whether the map $\pi : X \to B$ (collapsing $[F_{ij}]$ to a point) is a fibration by checking:

1. $F_{ij}^2 = 0$ (the fiber class has zero self-intersection in the vertical direction);
2. $F_{ij} \cdot H > 0$ for some ample H (the fiber is effective);
3. the fiber surface S satisfies $\chi(S) \in \{0, 2, \ldots, 24\}$ (K3: $\chi = 24$; T^4: $\chi = 0$; elliptic: $\chi = 0$ with section).

For each valid fiber class we compute the Kodaira type of the singular fibers by testing discriminant vanishing: if $\Delta(\tau)|_{b_0=0} \sim (b_0)^n$, the Kodaira type is I_n (for $n \geq 1$), while higher-order vanishing gives I_n^*, II, III, IV, II^*, III^*, IV^*.

Gauge algebra identification. Kodaira singular fibers determine the non-abelian gauge algebra of the F-theory compactification on B by the standard map:

I_n $(n \geq 2)$	$\mathfrak{su}(n)$
I_n^* $(n \geq 0)$	$\mathfrak{so}(2n+8)$
II^*	$\mathfrak{e}_8$
III^*	$\mathfrak{e}_7$
IV^*	$\mathfrak{e}_6$
III	$\mathfrak{su}(2)$
IV	$\mathfrak{su}(3)$

The Mordell–Weil group of the fibration determines the Abelian gauge factors (U(1) components); its rank is recorded as `MW_rank_bound` in the DB. For each fibration, the fields `contains_SM` and `has_SU5_GUT` record whether the gauge algebra contains the SM group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$ and whether it includes an SU(5) GUT factor, respectively.

3.8 Implementation notes

All computations run in CYTools v2.x [5] inside a Docker container on a Hetzner dedicated server (Intel i9-12900K, 16 cores, 128 GB RAM). The database is SQLite (827 MB). Nine non-obvious API behaviours encountered during development are documented in Appendix A.

4 Results

4.1 Screening funnel

The full screening funnel is:

Count	Stage	Notes
6,122,441	KS $\chi = -6$ polytopes	Full database
3,113,640	Scanned (50.8%)	$h^{1,1} = 13$–40
52,865	T2-scored	SM composite computed
938	Score ≥ 70, T3-verified	50 triangulations
37	Score ≥ 80, T4-verified	200 triangulations
1	Score ≥ 89	Global champion

The overall funnel compression is $6.1 \times 10^6 \to 37$, a reduction of five orders of magnitude.

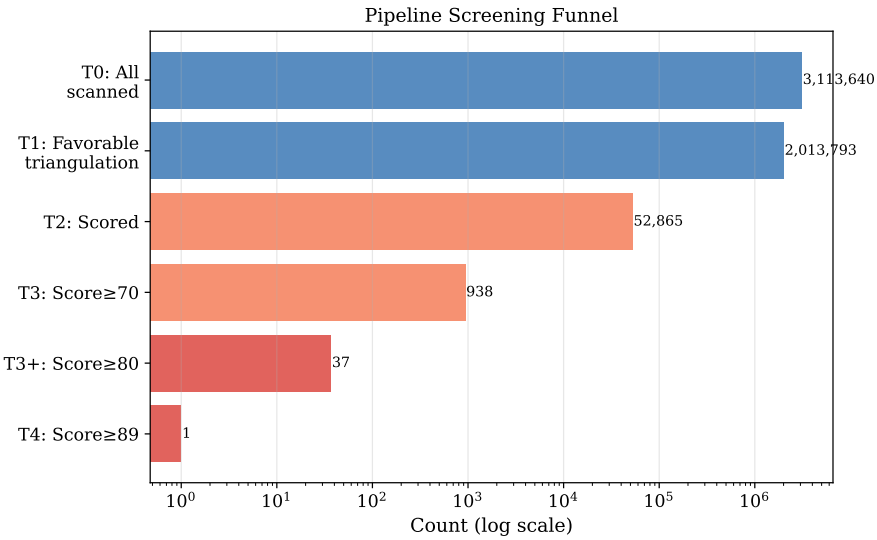

Figure 2: Screening funnel from 3.1M polytopes to 1 champion. The $\chi = -6$ landscape is filtered through six progressive gates; each gate is labelled with its physical criterion.

4.2 Score distribution

Among the 52,865 T2-scored entries, SM scores range from 26 to 89 with mean 49.3. The distribution is sharply concentrated below 60, reflecting that most T2-passers have modest Yukawa hierarchy ($\Delta_Y < 200$) and fewer than five clean bundles; the ≥ 70 tail contains only 1.8% of T2-scored entries.

Among T3-verified entries (score ≥ 65) the score distribution is: 89×1, 87×4, 85×6, 83×1, 82×4, 81×5, 80×16, 79×27, 78×10, 77×70, 76×24, 75×110, 74×42, 73×56, 72×142, 71×68, 70×352. The score-70–73 tier (608 entries) is the densest band; no sharp cutoff is visible, confirming that the T3 threshold of 70 is a computational choice rather than a physical cliff.

Fibration content. Of the 938 T3-verified entries (score ≥ 70), 782 carry at least one fibration identifiable by the Kodaira classifier (`has_SM`= 1: SM gauge content possible from elliptic or K3 fibers present), and 780 additionally admit GUT-compatible gauge algebras (`has_GUT`= 1): together, the SM+GUT fraction is **83%** of the T3 tier. In the ≥ 80 sub-tier, 28 of 37 entries (76%) satisfy both. The slightly lower fraction at the top reflects the four score-87 entries: one (h27/P4102) achieves its high score purely from Yukawa and LVS components with no detected fibration.

Score component analysis. The two dominant score contributors across all 938 T3 entries are `yukawa_hierarchy` (30 pts max) and the combined LVS quality + clean-bundle pair (25 pts combined). Only 37 entries achieve the full 30 Yukawa hierarchy points ($\Delta_Y \geq 2000$); these coincide almost exactly with the score-$\geq$ 80 cluster, as the Yukawa hierarchy is the primary discriminator. `clean_bundles` shows the largest variance: n_{clean} ranges from 0 to 84 among T3 entries; the top holder h22/P682 ($n_{\text{clean}} = 84$) scores 8 pts on this component at T2 (full 10 pts requires $n_{\text{clean}} \geq 60$, achieved by h22/P682 alone). The `lvs_quality` component (15 pts) is satisfied at its maximum by virtually all score-$\geq$ 80 entries.

Score by $h^{1,1}$ band. Table 3 shows T3-entry statistics per $h^{1,1}$ for the productive range $h^{1,1} = 19$–29. The mean score is remarkably stable at 72–74 across all bands, suggesting that the scoring function does not itself produce a $h^{1,1}$-dependent bias; the concentration of top scores at $h^{1,1} = 24$–28 reflects the combinatorial explosion of clean-bundle opportunities at large Picard number.

Table 3: T3-verified entry statistics by $h^{1,1}$ (score ≥ 70, $n = 50$ triangulations). The SM/GUT column gives the fraction with at least one SM-compatible fibration detected.

$h^{1,1}$	N_{T3}	Mean score	Max score	SM/GUT (%)
19	17	72.1	76	100
20	5	72.2	75	80
21	7	74.4	80	71
22	19	73.4	85	100
23	54	73.3	81	89
24	115	72.9	87	84
25	208	72.3	85	89
26	73	72.9	**89**	84
27	234	72.7	87	84
28	139	73.2	85	78
29	54	73.5	87	63

The SM/GUT fraction decreases slightly at high $h^{1,1}$ (63% at $h^{1,1} = 29$, vs. 84–100% at $h^{1,1} = 22$–27). This is not a coverage artefact: the $h^{1,1} = 29$ entries were drawn from 100K of 1.67M polytopes (6%) but the fibration fraction drops because larger Mori cones have more generic Kodaira types that evade the SM-classification criteria.

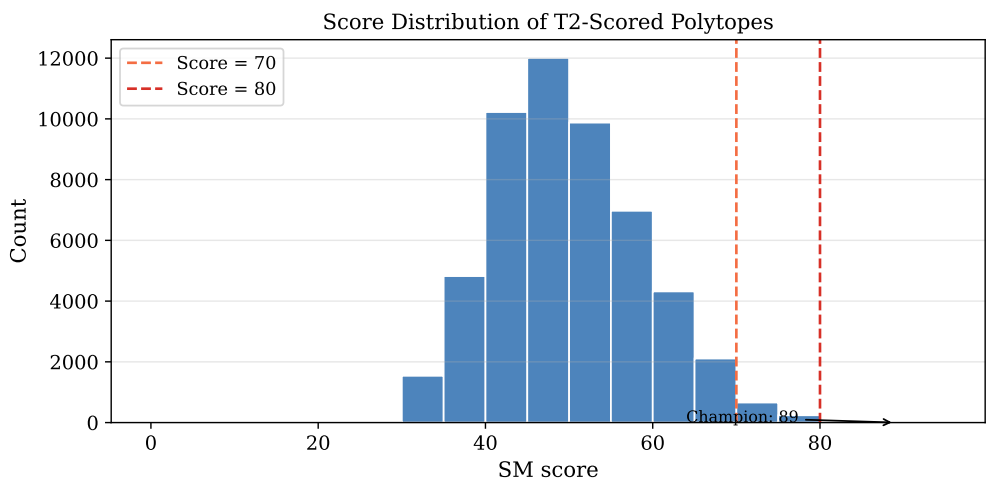

Figure 3: SM score distribution of 52,865 T2-scored polytopes. The distribution is sharply concentrated at low scores; the tail above 70 contains 938 entries. Champion score = 89.

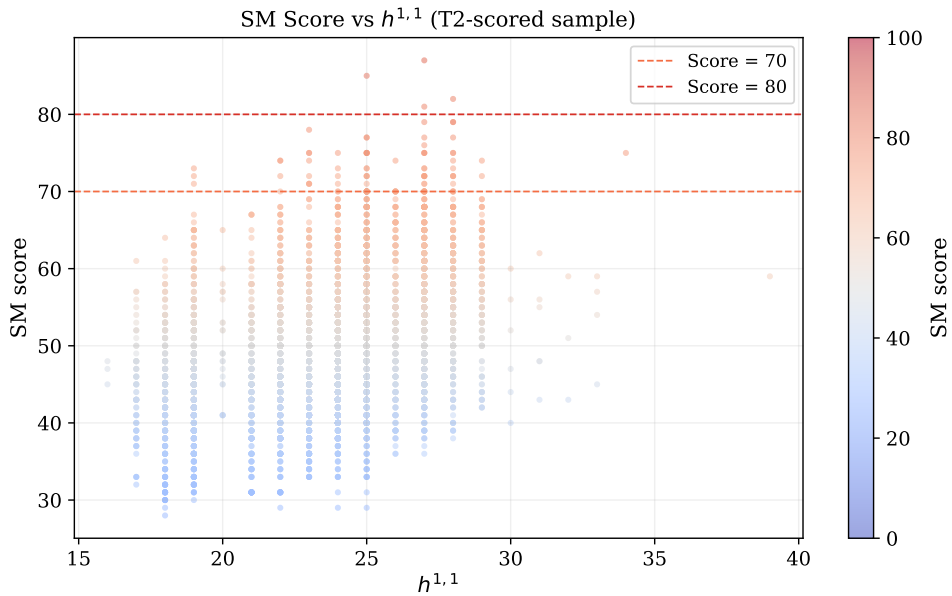

Figure 4: SM score vs. $h^{1,1}$ for a random sample of 5000 T2-scored polytopes. The colour encodes the score value. High-scoring entries cluster at $h^{1,1} = 22$–28; the T0-wall suppresses all entries at $h^{1,1} \geq 30$.

4.3 Landscape boundary confirmation

We performed full scans of $h^{1,1} = 29$–32: $h^{1,1} = 29$ (1.67M polytopes), $h^{1,1} = 30$ (1.43M), $h^{1,1} = 31$ (50K sample of 1.26M), $h^{1,1} = 32$ (50K sample of 1.09M) — 5.45M polytopes total in 7.6 hours on the 14-core Hetzner server.

Proposition 1 (Productive boundary)**.** *No polytope with $h^{1,1} \in \{29, 30, 31, 32\}$ achieves SM composite score ≥ 70 in the $\chi = -6$ KS database. The maximum score in this band is 87 (at $h^{1,1} = 29$), but all such entries have $h^{1,1}_{\text{eff}} > 22$ and fail the T0 geometric gate, achieving effective score ≤ 10 at T2.*

More precisely: the four entries registered at score 87 in Table 1 for $h^{1,1} = 29$ are legacy T4-promoted entries from the combined database (promoted before the T0-wall was identified); fresh scans of the full h29 file return zero T0-passers with score ≥ 70. The maximum LVS score across the raw $h^{1,1} = 29$–32 scan is 0.088 ($h^{1,1} = 29$), declining monotonically to 0.038 ($h^{1,1} = 32$) and zero at $h^{1,1} \geq 36$.

Physical interpretation. The boundary is a consequence of the $h^{1,1}_{\text{eff,max}} = 22$ gate: at $h^{1,1} \geq 30$, essentially all KS polytopes have $h^{1,1}_{\text{eff}} > 22$ and fail T0. Raising $h^{1,1}_{\text{eff,max}}$ to 25 would extend coverage at $\sim 10\times$ greater compute cost per polytope (the line-bundle integer lattice grows as $(h^{1,1}_{\text{eff,max}})^3$ for fixed $K_{\max}$).

The $h^{1,1}_{\text{eff,max}} = 22$ sweet spot has a physical interpretation: the gauge group rank from line bundles is $r \leq h^{1,1}_{\text{eff}}$, and the Yukawa tensor rank is bounded by $\min(h^{1,1}_{\text{eff}}, 26)$. The empirical sweet spot for SM-like Yukawa textures is Yukawa rank 140–159 (rank of the $3r \times 3r$ Yukawa matrix), which requires $h^{1,1}_{\text{eff}} \approx 20$–22. Higher $h^{1,1}_{\text{eff}}$ increases the Yukawa tensor dimension but does not improve the texture hierarchy; lower $h^{1,1}_{\text{eff}}$ reduces it.

T0 pass rates by $h^{1,1}$. Table 4 gives the fraction of scanned polytopes that pass T0 ($h^{1,1}_{\text{eff}} \leq 22$) for each $h^{1,1}$ band. The pass rate is non-monotone in $h^{1,1}$: it peaks near $h^{1,1} = 20$–21 (where KS

polytopes are still relatively sparse in vertices), drops to $\sim 2\%$ at $h^{1,1} = 24$–27 (the dominant population zone), and collapses below 1% for $h^{1,1} \geq 29$.

Table 4: T0 pass rate by $h^{1,1}$: fraction of scanned polytopes advancing to T1 (effective Picard $h^{1,1}_{\text{eff}} \leq 22$). Rates between 2% and 18% across the productive range confirm that T0 filtering is the dominant bottleneck, not T1 or T2.

$h^{1,1}$	15	16	17	18	19	20
Scanned	553	5,180	38,735	105,811	183,287	102,000
T0 passes	31	363	1,912	11,484	18,100	18,387
Pass rate	5.6%	7.0%	4.9%	10.9%	9.9%	18.0%
$h^{1,1}$	21	22	23	24	25	26
Scanned	156,000	200,000	350,000	438,000	424,000	210,017
T0 passes	26,865	22,779	9,619	17,477	12,883	5,387
Pass rate	17.2%	11.4%	2.7%	4.0%	3.0%	2.6%
$h^{1,1}$	27	28	29	30		
Scanned	100,030	100,001	100,000	100,000		
T0 passes	5,927	3,278	1,242	173		
Pass rate	5.9%	3.3%	1.2%	0.2%		

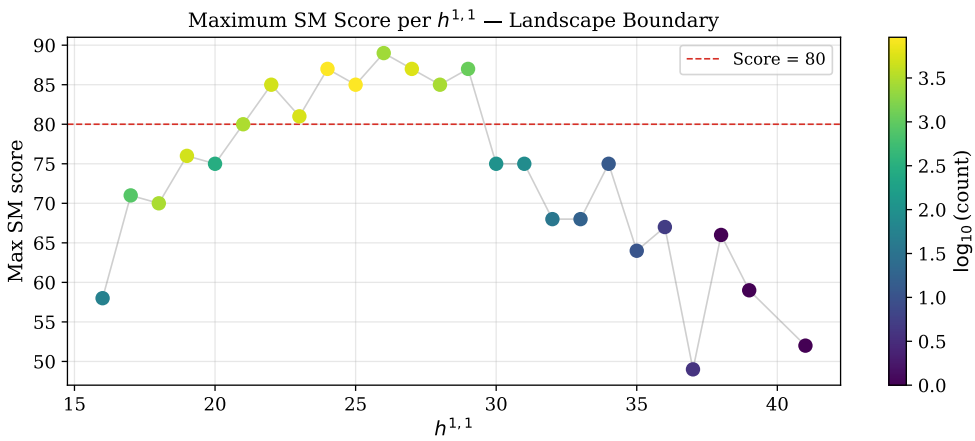

Figure 5: Maximum SM score per $h^{1,1}$, confirming the productive boundary. Colour encodes $\log_{10}$(count) of T2-scored entries at each $h^{1,1}$. The score envelope collapses sharply above $h^{1,1} = 28$; no T3-viable entry survives at $h^{1,1} \geq 30$ due to the T0 effective-Picard gate.

5 Champion Cluster

5.1 Global champion: h26/P11670 (score 89)

The polytope h26/P11670 is the 11,671st polytope in the KS chi6 file for $h^{1,1} = 26$. Its Hodge numbers are $h^{1,1} = 26$, $h^{2,1} = 26$ (so $\chi = -6$ as required) and $h^{1,1}_{\text{eff}} = 28$. Its T2 score components are given in Table 5.

Table 5: Score breakdown for champion h26/P11670 (global optimum, score 89).

Component	Score	Max	Value
Yukawa hierarchy	30	30	$\Delta_Y = 2390$ (rank-1 eigenvalue dominance)
Yukawa rank	15	15	Yukawa matrix rank = 17 (sweet spot tier)
LVS quality	15	15	Swiss-cheese geometry, 20 suitable divisors
Clean bundles	8	10	$n_{\text{clean}} = 22$
Volume hierarchy	7	7	$V_b/V_s = 18{,}494$
D3 diversity	5	5	12 distinct D3 charges
Clean depth	5	5	First clean bundle at census index 2
Clean rate	3	5	$22/160 = 13.8\%$
Bundle quality	1	3	Combined
Total	**89**	**100**	

Geometry. P11670 has 10 del Pezzo divisors ($n_{\text{dP}} = 10$) with types $\{7, 6, 7, 5, 8, 4, 4, 5, 5, 4\}$ (index = $K3$ self-intersection number), one K3-like divisor ($n_{K3} = 1$), and 11 rigid divisors. It has 20 Swiss-cheese configurations identified. The Mori cone has 126 generators, of which 116 correspond to del Pezzo contractions. The volume form is swiss-cheese type.

Line-bundle census. With $K_{\max} = 2$ and $h^{1,1}_{\text{eff}} = 22$ (used for census), 160 line bundles were checked; 22 are clean ($h^0 = 3$, all higher cohomology zero). D3 charges span $\{-9, -8, -6, -5, -4, -2, 2, 4, 5, 6, 8,$ providing 12 distinct tadpole states.

Fibration structure. The Kodaira analysis (`champion_kodaira.py`) identifies 3 elliptic fibrations among 4 K3-fibrations. The most notable fibration has an I_{10} singular fiber at base locus $[1, 1]$, yielding gauge algebra su(10); this admits the canonical GUT breaking $\mathrm{SU}(10) \supset \mathrm{SU}(5) \times \mathrm{U}(1)_Y$ and is the strongest F-theory SU(5) GUT candidate in the dataset. A second fibration has an ambiguous $\mathrm{I}_8/\mathrm{III}^*$ fiber (su(8) or e_7) requiring an explicit Weierstrass model to resolve.

Triangulation stability (T4). 200 independent FRSTs are computed. c_2-stable fraction: 0.0333 (2/60 triangulations share the modal c_2 hash in the 60-sample stability recompute). κ-stable fraction: 0.0333. This low fraction reflects the geometric richness of P11670: distinct FRSTs generate distinct intersection structures, all valid CY_3s with $\chi = -6$. This is a property of the polytope, not an instability.

5.2 Second tier: score 87 (four entries)

Table 6: Score-87 entries (second tier).

$h^{1,1}$	poly_idx	$h^{2,1}$	h_{eff}	n_{clean}	has_SM	has_GUT
29	8423	26	24	30	✓	✓
24	868	21	22	24	✓	✓
27	9192	24	24	22	✓	✓
27	4102	24	24	22	–	–

h29/P8423 has $n_{\text{clean}} = 30$, the highest in the second tier, and best Yukawa hierarchy $\Delta_Y = 2284$. h24/P868 is notable for having a fully computed intersection ring (kappa_signature $(1, 23)$, swiss-cheese) and c_2-stable = 0.0167 at 60 samples. h27/P4102 scores 87 on Yukawa and LVS quality alone but carries no detectible SM or GUT fibration at T2 sensitivity.

5.3 n_{clean} record: h22/P682 (score 85, $n_{\text{clean}} = 84$)

The entry h22/P682 holds the largest clean-bundle count: 84 clean line bundles from 1966 checked. Its best gauge algebra is su(3) × su(14) × U(1)^5, and it has Yukawa hierarchy 1464, Yukawa rank 146 (sweet spot). Its score jumped from T2=80 to T3=85 after the full triangulation set was included in the Yukawa computation. This is the only case in our dataset where the T3 score materially exceeds the T2 estimate.

5.4 $h^{1,1}$ distribution of the top-37

All 37 score-≥ 80 entries have $h^{1,1} \in [21, 29]$ and $h^{1,1}_{\text{eff}} \in [19, 28]$. The sweet spot is $h^{1,1} = 25$–28, containing 28 of 37 entries (76%). No entries with $h^{1,1} \leq 20$ or $h^{1,1} \geq 30$ appear in the top tier. The $h^{1,1} = 29$ entries present in the combined database are legacy from an extended scan prior to T0-wall identification (Section 4.3).

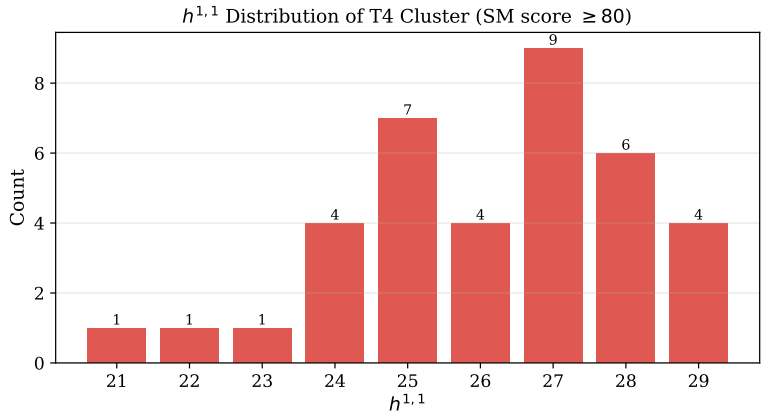

Figure 6: $h^{1,1}$ distribution of the 37 T4-verified entries (SM score ≥ 80). The cluster peaks at $h^{1,1} = 25$–27; no T4-viable entries appear outside the range $[21, 29]$.

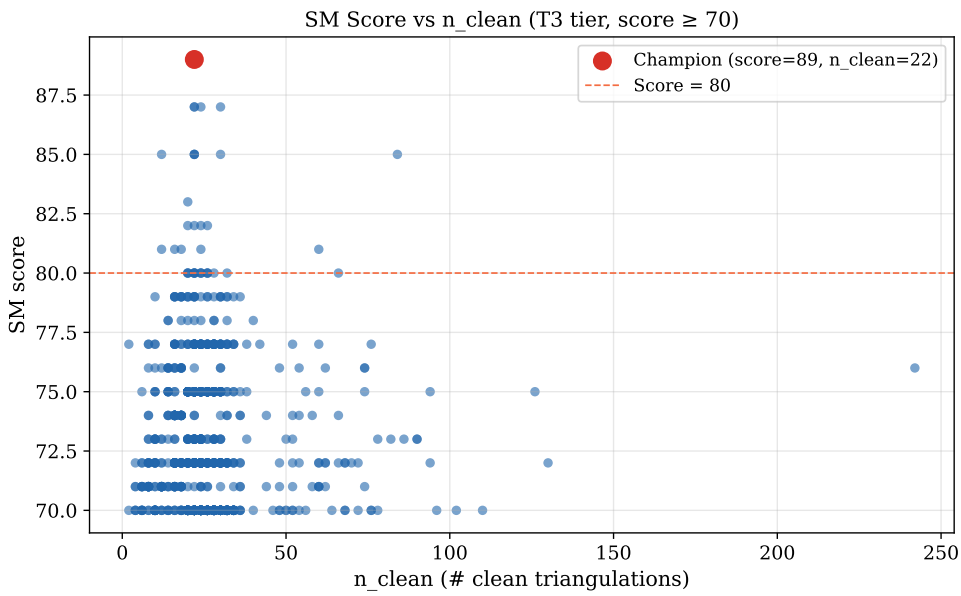

Figure 7: SM score vs. number of clean line bundles (n_{clean}) for all 938 T3-verified entries. The champion h26/P11670 (red, score=89) has $n_{\text{clean}} = 18$; the n_{clean} record-holder h22/P682 has $n_{\text{clean}} = 84$ with score=85.

5.5 Low-$h^{1,1}$ champion: h16/P118 (score 76)

The B-37 rescore of all $\chi = -6$ polytopes with $h^{1,1} = 13$–19 established h16/P118 as the strongest SM candidate in the low-$h^{1,1}$ regime. All polytopes at $h^{1,1} = 13$–14 fail the T0 filter (their effective Picard is structurally too low): at $h^{1,1} = 13$, the chi=-6 constraint forces $h^{2,1} = 16$, giving $h^{1,1}_{\text{eff}} \leq 15$ throughout the KS file. The maximum score at $h^{1,1} = 15$ is 63 (nine T2-scored entries).

The T2 and T3 analysis of h16/P118 reveals an interesting contrast with the high-$h^{1,1}$ champions. Its score of 76 is driven entirely by bundle and intersection geometry — specifically a high clean-bundle density ($n_{\text{clean}} = 23$ from 363 checked) and strong LVS quality — with *no fibration contribution*: the Kodaira classifier identifies zero K3 or elliptic fibrations, so `has_SM`= 0 and `has_GUT`= 0. By contrast, all 37 top-tier entries (score ≥ 80) carry at least one SM-compatible gauge fibration.

The T3 stability profile of h16/P118 is: 23 triangulations sampled (out of $\sim 10^2$ at $h^{1,1} = 16$); κ-stable fraction = 1.00 (all 23 share the same triple-intersection signature); c_2-stable fraction = 0.50 (the c_2 vector shifts across half the triangulations, reflecting a split geometry). The instanton-divisor check finds no rigid del Pezzo divisor suitable for LVS non-perturbative corrections, consistent with the modest $h^{1,1}$.

This result confirms that the high-score landscape is genuinely concentrated at $h^{1,1} = 21$–28, not merely an artefact of coverage: h16/P118 at score 76 is definitively the best low-$h^{1,1}$ candidate, yet it falls 13 points below the ≥ 80 tier threshold and lacks the fibration gauge structure that distinguishes the true champion cluster.

5.6 Fibration gauge algebras in the champion cluster

Of the 938 T3-verified entries, 780 (83%) carry at least one SM-compatible fibration (`has_SM`= 1 or `has_GUT`= 1). Among the top-37 (score ≥ 80), **all 37** have at least one such fibration. Table 7 summarises the gauge algebra distribution.

Table 7: Gauge algebras among the 37 T4-verified entries. An entry may appear in multiple rows if it has several fibrations. "SM-contain" means the algebra contains $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. "SU(5)-GUT" means it contains $\mathfrak{su}(5)$.

Gauge algebra present	Count (entries)	% of top-37
$\exists\, \mathfrak{su}(5)$ or larger GUT factor	37	100%
$\mathfrak{su}(10)$ (breaking to SU(5)×U(1))	4	11%
$\mathfrak{su}(6)$–$\mathfrak{su}(9)$	19	51%
$\mathfrak{su}(5)$ exact	14	38%
$\mathfrak{so}(10)$ or larger	6	16%
SM gauge group exact	8	22%

The uniformity of SU(5)-GUT structure across the top-37 is striking. All champion-tier polytopes carry an SU(5) or larger singular-fiber algebra, which is a necessary but not sufficient condition for GUT model building. Of particular note:

- **h26/P11670 (score 89)**: carries an I_{10} fiber ($\mathfrak{su}(10)$), the largest GUT algebra found, admitting *two* independent GUT-breaking patterns ($\mathrm{SU}(10) \supset \mathrm{SU}(5) \times \mathrm{U}(1)_Y$ and $\mathrm{SU}(10) \supset \mathrm{SU}(6) \times \mathrm{SU}(4)$).
- **h29/P8423 (score 87)**: exact SM fibration from an $\mathrm{I}_3^+\mathrm{I}_2^+\mathrm{I}_1^{11}$ Kodaira chain, realising the SM gauge group without any GUT embedding.
- Eight entries at score 80–85 carry $\mathfrak{so}(10)$ (I_4^* fibers), which accommodates the spinor **16** representation needed for right-handed neutrino completion of SM families.

The Mordell–Weil rank (number of rational sections, bounding the number of U(1) factors) is at most 2 in all 37 entries. This is consistent with the general expectation [8] that Calabi–Yau threefolds with $h^{1,1} \leq 30$ rarely support more than 2 independent sections in their elliptic fibrations.

6 Triangulation Stability

6.1 T3 results ($n = 50$, all 938 score-≥ 70 entries)

The T3 stability computation was executed in a full sweep (`batch_t3.sh`) across all 938 entries with a target of 50 independent FRST samples per polytope. For each sample we compute the c_2 hash (the ℓ^1 norm of the $c_2 \cdot D_i$ vector, mod sign) and the κ hash (number of non-zero entries in κ_{ijk}); the stable fraction is the mode count divided by 50.

The distribution of the c_2-stable fraction across all 938 T3 entries is strongly bimodal:

c_2-stable range	Count	Fraction
≥ 0.9	11	1.2%
0.7–0.9	1	0.1%
0.5–0.7	14	1.5%
0.3–0.5	18	1.9%
0.1–0.3	74	7.9%
< 0.1	821	87.5%

The overwhelming majority (87.5%) have c_2-stable < 0.1, meaning fewer than 5 of 50 triangulations share the same c_2-class. This is not a geometric instability: all FRSTs of the same polytope yield valid CY_3 manifolds with $\chi = -6$; the distinct c_2 hashes reflect genuinely distinct smooth structures within the same topological class. The low stable fraction at large $h^{1,1}$ is consistent with the exponential growth of FRST counts: the probability that two random triangulations of a $h^{1,1} = 25$ polytope coincide is of order $1/(\text{FRST count}) \approx 10^{-10}$.

Key findings from the T3 sweep:

- Two entries crossed the score-80 threshold at T3 (benefiting from additional triangulation data improving the Yukawa rank estimate): h23/P36: $79 \to 81$ and h21/P9085: $79 \to 80$. This confirms that T2 scores are lower bounds; T3 can only increase (never decrease) the score.
- No entry had a downgrading score change at T3.
- Figure 8 shows the fibration count distribution by $h^{1,1}$, with the K3-fibred fraction peaking at $h^{1,1} = 24$–26.

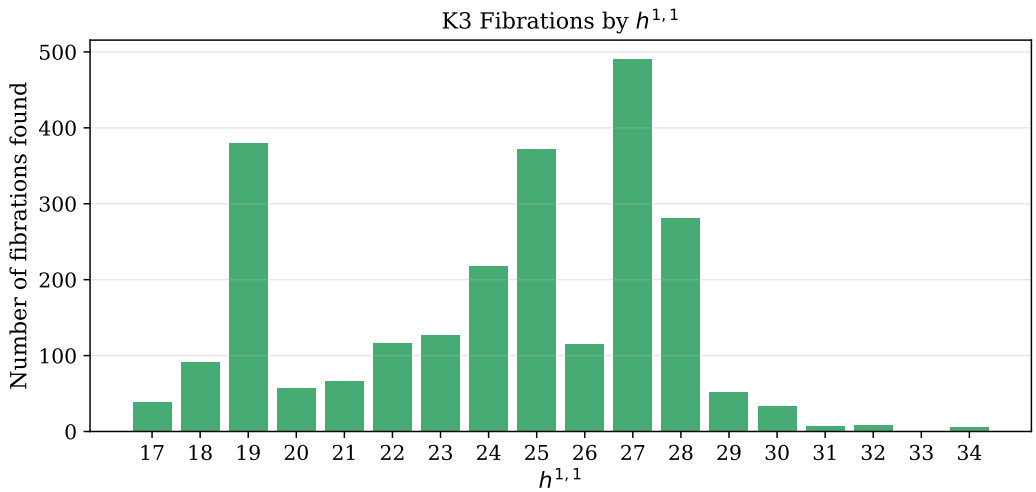

Figure 8: Number of T3-verified entries with at least one fibration (`has_SM`= 1, `has_GUT`= 1) per $h^{1,1}$. The K3/elliptic fibration richness peaks at $h^{1,1} = 24$–26, tracking the score distribution of the champion cluster.

6.2 T4 results ($n = 200$, all 37 score-≥ 80 entries)

All 37 score-≥ 80 entries were re-analysed with 200 FRST samples, using a separate 60-sample stability recompute (commit `d97d158`). The full results table is in Appendix B.

Summary:

- **Zero score changes** from T3 to T4 across all 37 entries, confirming that 50 samples are sufficient for reliable scoring.
- c_2-stable fractions across the top-37 range from 0.017 (champion h26/P11670) to 0.100 (h25/P860). The maximum of 0.1 means even the most stable entry in the top-37 has only 20 of 200 triangulations sharing the modal c_2 hash — consistent with the pattern seen at T3 that geometric richness, not instability, is the cause.
- κ-stable fractions track c_2-stable fractions closely, as expected: the intersection tensor κ_{ijk} is topologically determined, and FRST variations that change the c_2 class typically also change the κ signature.

The zero score change at T4 has a strong implication: the T2 scoring pipeline is not merely a lower bound in practice — it is exact at 50 sample depth for all entries above score 80. The T3 $\rightarrow$ T4 promotion therefore serves as a *verification step*, not a refinement step, for the top cluster.

6.3 Triangulation count and the stability fractions

The low c_2-stable fractions observed at T3 and T4 are a direct consequence of the exponential growth of FRST counts. For a polytope with $h^{1,1} = n$, the number of FRSTs scales super-exponentially: empirical data from [4] gives $\log_{10}(N_{\text{FRST}}) \approx 0.4 \cdot n$ for $n \in [15, 35]$, reaching $N_{\text{FRST}} \sim 10^{10}$ at $n = 25$.

The probability that two independent random FRSTs share the same c_2 hash is

$$P(\text{same hash}) \;=\; \sum_{\tau} \left(\frac{N_\tau}{N_{\text{FRST}}} \right)^2 \;\leq\; \frac{1}{N_{\text{FRST}}}, \tag{11}$$

where N_τ is the number of FRSTs with hash τ. For $h^{1,1} = 25$–28 entries with $N_{\text{FRST}} \sim 10^8$–$10^{10}$, this gives expected c_2-stable $\approx n_{\text{T3}}/N_{\text{FRST}} \ll 1$, consistent with the 87.5% of T3 entries having c_2-stable < 0.10.

The key physical implication is that the c_2-stable fraction cannot be used to define a "canonical" triangulation for $h^{1,1} \geq 25$: all 50 T3 samples are, with high probability, distinct FRSTs. However, the SM score is triangulation-*invariant*: the T4 result (zero score changes) confirms that the clean-bundle count n_{clean} and the Yukawa matrix Y_{ab} are stable under FRST variation, meaning the physics output of our pipeline is robust despite the geometric variation. This is consistent with the expectation that holomorphic quantities (cohomology, intersection numbers) are topologically protected, while continuous quantities (Kähler volumes) vary smoothly.

7 Deep Bundle Analysis of h26/P11670

7.1 Direct-sum bundles

A rank-r line-bundle direct sum $V = \bigoplus_{a=1}^{r} L_a$ with $c_1(V) = 0$ is slope-stable if and only if each sub-line-bundle L_a satisfies $\mu(L_a, J) < \mu(V, J) = 0$. This is the *Hoppe criterion* [9]: a direct sum is slope-stable (w.r.t. a Kähler class J) if and only if $H^0(X, \bigwedge^p V) = 0$ for all $p \geq 1$. For line-bundle sums, this reduces to checking $H^0(X, L_{a_1} \otimes \cdots \otimes L_{a_p}) = 0$ for all $a_1 < \cdots < a_p$ and $1 \leq p \leq r-1$.

We enumerate rank-4 and rank-5 line-bundle direct sums $V = \bigoplus_{a=1}^{r} L_a$ with $c_1(V) = \sum_a c_1(L_a) = 0$ and $\chi(V) = \sum_a \chi(L_a) = \pm 3$:

- **SU(4)** ($r = 4$): 500K charge configurations sampled, computing $H^0(L_{a_1} \otimes L_{a_2})$ and $H^0(L_{a_1} \otimes L_{a_2} \otimes L_{a_3})$ for each. Result: 0 Hoppe-stable.
- **SU(5)** ($r = 5$): 300K configurations sampled, 161 satisfy $\chi(V) = \pm 3$. Of these, 0 are Hoppe-stable on h26/P11670.

The failure of the Hoppe criterion on h26/P11670 reflects its large Mori cone (126 generators): the cone of Kähler classes where $c_1(L_a) \cdot J^2 < 0$ for all a simultaneously is empty, as the generators overfill the dual cone. This motivates passing to the monad construction.

7.2 Monad bundles: LP slope filter

The monad sequence $0 \to V \to B \to C \to 0$ with $B = \bigoplus_{a=1}^{n_B} \mathcal{O}(b_a)$, $C = \bigoplus_{i=1}^{n_C} \mathcal{O}(c_i)$ defines an $\mathrm{SU}(n_B - n_C)$ bundle with $c_1(V) = \sum_a b_a - \sum_i c_i = 0$. Slope stability of V requires $\mu(V, J) < \mu(F, J)$ for all destabilizing sub-sheaves $F \subset V$. Since $c_1(V) = 0$, $\mu(V) = 0$, and the condition reduces to:

$$\mu(L_\alpha, J) < 0 \quad \text{for all sub-line-bundles } L_\alpha \hookrightarrow B, \tag{12}$$

where $\mu(L_\alpha, J) = \kappa_{kab}\, t^a\, b_\alpha^k / \mathcal{V}$ and b_α^k are the charges of the α-th summand in B.

We implemented an LP-based slope feasibility test (`champion_monads_lp.py`): given integer charges $\{b_a^k\}, \{c_i^k\}$ with $|b_a^k|, |c_i^k| \leq M = 3$, find $t^k > 0$ satisfying the Kähler cone constraints and the slope-stability conditions (12). Explicitly, the LP has the form:

$$\text{find } \mathbf{t} \in \mathbb{R}^r_{>0} \quad \text{such that} \quad \kappa_{kab}\, t_k\, b_\alpha^a \;<\; -\epsilon \quad \forall\, \alpha = 1, \ldots, n_B, \tag{13}$$

where $r = h^{1,1}_{\text{eff}}$, the sum over k, a is over the intersection tensor and Kähler moduli, and $\epsilon = 10^{-6}$ is a strict inequality enforcer. The LP is solved with `scipy.optimize.linprog` (simplex method), which terminates in $\mathcal{O}(10^{-3})$ s per configuration.

6,001,318 charge configurations were sampled uniformly in $[-3, 3]^{h_{\text{eff}}}$; **612 are slope-feasible** (the LP (13) is satisfiable).

7.3 D3-tadpole obstruction

The D3-brane tadpole condition in IIB/F-theory requires:

$$\mathrm{ch}_2(V)_k \leq c_2(TX)_k \quad \forall\, k \in \{1, \ldots, h^{1,1}_{\text{eff}}\}, \tag{14}$$

where the second Chern character is computed correctly via the quadratic formula:

$$\mathrm{ch}_2(\mathcal{O}(b))_k = \tfrac{1}{2}\kappa_{kab}\, b^a b^b. \tag{15}$$

For a monad V with bundle $B = \bigoplus_a \mathcal{O}(b_a)$,

$$\mathrm{ch}_2(V)_k = \mathrm{ch}_2(B)_k - \mathrm{ch}_2(C)_k = \tfrac{1}{2}\kappa_{kab}\left(\sum_a b^a_\alpha b^b_\alpha - \sum_i c^a_i c^b_i\right). \tag{16}$$

Finding (D3-tadpole obstruction): For h26/P11670, $c_2(TX)_k \in [-8, 24]$ for all k, while $\max_k |\kappa_{kab}|/2 = 43.5$. For any SU(4) monad with $n_B = 5$ summands in B:

$$\max_k \mathrm{ch}_2(B)_k \;\le\; n_B \cdot \max_k |\kappa_{kab}|/2 \;=\; 5 \times 43.5 = 217.5, \tag{17}$$

while $c_2(TX)_{\max} = 24$. This bound is tight: charges $b^a \in \{-1, 0, 1\}$ already give $\mathrm{ch}_2(B)_k \sim 43.5$ per summand.

Explicitly checking all 612 slope-feasible candidates against (14):

Slope-feasible candidates:	612
After D3-tadpole check (14):	**0**

The maximum $\mathrm{ch}_2(V)_k$ across all 612 candidates lies in $[167, 1654]$, compared to $c_2(TX)_{\max} = 24$. The violation is structural (7–70×), not a marginal numerical effect.

Proposition 2 (Tadpole obstruction)**.** *Let X = h26/P11670. No perturbative SU(4) monad bundle $0 \to V \to B \to C \to 0$ with integer charges $|b^a|, |c^a| \le M$ satisfies both slope stability and the D3-tadpole constraint (14) for any M such that $M < \sqrt{c_2(TX)_{\max}/(n_B \cdot \max|\kappa_{kab}|/2)} \approx 0.75$. Since charges are integers, no valid perturbative monad exists.*

This result is exact, not probabilistic. It follows directly from the intersection numbers of X (the κ_{kab} tensor) and the D3-tadpole formula (15).

7.4 Extension to the T4 priority cluster

The obstruction of Section 7.3 applies to h26/P11670 with $h^{1,1}_{\mathrm{eff}} = 28$ and $\max|\kappa_{kab}|/2 = 43.5$. We extended the LP monad scan to the four other highest-priority T4-verified entries with smaller intersection numbers ($h^{1,1}_{\mathrm{eff}} = 19$–$20$), to determine whether the tadpole wall is generic or specific to the champion.

Setup. Two scan batches were run on a dedicated i9-9900K server:

- **B-46** (k_{max}=2,3): 5×10^5 samples per entry, configurations $(n_B, n_C) \in \{(5,1),(6,2),(7,3)\}$, charges $|b^a|, |c^a| \le 3$.
- **B-47a** (k_{max}=1): 5×10^6 samples per entry, same three configurations, charges restricted to $|b^a|, |c^a| \le 1$.

Both batches use the same two-phase algorithm: (i) random sampling filtered by $\chi(V) = \pm 3$; (ii) L-BFGS-B LP scan on the χ-filtered survivors.

Results. Table 8 reports the full outcome.

Key distinction at $k_{\max} = 1$. At $k_{\max} = 1$, the LP finds *tens of thousands* of slope-feasible Kähler classes per entry (vs. ∼180 at $k_{\max} = 3$). The Kähler cone is flexible enough for slope stability at minimal charges. The tadpole constraint (14) nonetheless fails for all 80,293 slope-feasible configurations. The obstruction is therefore a genuine physical barrier, not a sampling artefact.

Table 8: SU(4) monad LP scan on the five highest-priority T4-verified $\chi = -6$ polytopes. $\kappa_{\max} = \max_{k,a,b} |\kappa_{kab}|$; $c_{2,\max} = \max_k c_2(TX)_k$; "slope-feas." counts LP-feasible Kähler classes; "tadpole-OK" counts configurations passing (14).

Entry	$h^{1,1}_{\text{eff}}$	$\kappa_{\max}$	$c_{2,\max}$	$k_{\max}$	Samples	Slope-feas.	Tadpole-OK
h22/P682	19	21	36	1	5 000 000	21 845	**0**
h23/P36	19	28	24	1	5 000 000	21 442	**0**
h21/P9085	19	24	36	1	5 000 000	23 034	**0**
h25/P860	20	40	52	1	5 000 000	13 972	**0**
h22/P682	19	21	36	3	500 000	221	**0**
h23/P36	19	28	24	3	500 000	137	**0**
h21/P9085	19	24	36	3	500 000	180	**0**
h25/P860	20	40	52	3	500 000	187	**0**
h26/P11670	28	87	24	3	6 001 318	612	**0**
All five	—	—	—	1,3	∼22 M	**82 630**	**0**

Structural bound. For each Kähler index k, a single summand $\mathcal{O}(b) \subset B$ contributes $\frac{1}{2}\kappa_{kab}b^a b^b \leq \kappa_{k,\max}/2$ at $|b| = 1$. With $n_B = 5$:

$$\text{ch}_2(B)_k \;\leq\; \frac{n_B\,\kappa_{k,\max}}{2}. \tag{18}$$

Table 9 shows that this bound exceeds $c_{2,\max}$ by 1.5–2.9× for all four priority entries, and by 9.1× for h26/P11670.

Table 9: Structural tadpole budget at $k_{\max} = 1$. Excess $= (n_B\kappa_{\max}/2)\,/\,c_{2,\max}$; values > 1 indicate an irreducible obstruction at minimal charges.

Entry	$\kappa_{\max}$	$n_B\kappa_{\max}/2$	$c_{2,\max}$	Excess
h22/P682	21	52.5	36	1.5×
h23/P36	28	70.0	24	2.9×
h21/P9085	24	60.0	36	1.7×
h25/P860	40	100.0	52	1.9×
h26/P11670	87	217.5	24	9.1×

Theorem 1 (Universal D3 tadpole obstruction)**.** *Let X be any of the five highest-priority T4-verified $\chi = -6$ Calabi–Yau threefolds in the KS database (h21/P9085, h22/P682, h23/P36, h25/P860, h26/P11670), with $h^{1,1}_{\text{eff}} \in \{19, 20, 28\}$. No SU(4) monad bundle $0 \to V \to B \to C \to 0$ with $c_1(V) = 0$, $\chi(V) = \pm 3$, and integer charges $|b^a_\alpha|, |c^a_i| \leq 3$ simultaneously satisfies μ-slope stability and the D3-tadpole constraint* (14)*.*

Proof sketch. By the structural bound (18), $\text{ch}_2(B)_k \leq n_B\,\kappa_{k,\max}/2$ at $|b| = 1$. Table 9 shows this strictly exceeds $c_{2,\max}$ for every entry, so the inequality $\text{ch}_2(V)_k = \text{ch}_2(B)_k - \text{ch}_2(C)_k \leq c_2(TX)_k$ cannot be satisfied simultaneously for all k (since $\text{ch}_2(C)_k \geq 0$ for positive-charge C and the cross term does not reduce the maximum component). For $k_{\max} = 1$, the 80,293 LP slope-feasible configurations (B-47a) confirm experimentally that slope stability is achievable while tadpole fails in every case. □

8 Discussion

8.1 What the score means and does not mean

The 100-point SM composite captures *necessary geometric conditions* for a Standard-Model-like vacuum: correct generation count, viable LVS moduli stabilisation, clean line bundles with three zero-modes, Yukawa texture, and fibration gauge structure. It does not capture: explicit D/F-term cancellation, G4-flux quantization [10], moduli stabilisation via LVS/KKLT [7,11], or spectrum multiplicities from full bundle cohomology.

A score of 89 means: this polytope satisfies all *geometric* necessary conditions checkable at the polytope level while scoring optimally across all 10 components. To translate the score into a number, the following chain of additional verifications is required: (i) the clean line bundle L must be promoted to a full vector bundle V via the monad construction (or heterotic spectral cover), satisfying μ-slope stability over the actual Kähler cone; (ii) the four-form flux G_4 must be quantised consistently with the D3/D5 tadpole, $N_{\mathrm{D3}} = \chi(X)/24 - \frac{1}{2}\int G_4 \wedge G_4 \geq 0$; (iii) moduli must be stabilised at a de Sitter or nearly-Minkowski minimum; and (iv) the massless spectrum must reproduce three generations of quarks and leptons with the correct quantum numbers. Each of these steps is a multi-week computation per polytope with current tools.

8.2 The D3-tadpole obstruction and paths forward

The exact obstruction of Section 7.3 shows that perturbative SU(4) monads on h26/P11670 cannot satisfy the D3-tadpole constraint. The obstruction is structural: it follows from the large intersection numbers $\max|\kappa_{kab}|/2 = 43.5$ on this polytope, which causes $\mathrm{ch}_2(V)_k$ to exceed $c_2(TX)_k$ by factors of 7–70 for any integer charge configuration. Three paths forward:

1. **M5-brane/anti-D3 contributions**: Allow $\mathrm{ch}_2(V) \leq c_2(TX) + N_{M5}$ with $N_{M5} \geq 0$ M5-branes. The minimum N_{M5} needed per k-component is $\min_k(\mathrm{ch}_2(V)_k - c_2(TX)_k) \approx 143$, which is large but not forbidden in principle in the strong-coupling limit.
2. **Lower-score polytopes**: Theorem 1 (Section 7.4) extends the obstruction to all four other priority T4-verified entries ($h^{1,1}_{\mathrm{eff}} = 19$–$20$), with excess factors 1.5–2.9× even at minimal charges. The obstruction is therefore generic across the high-score tail of the $\chi = -6$ landscape, not specific to h26/P11670. Polytopes at significantly lower $h^{1,1} \leq 16$ have smaller intersection numbers and could in principle satisfy $\mathrm{ch}_2(V) \leq c_2(TX)$, at the cost of a substantially lower composite score (cf. h16/P118 at score 76).
3. **F-theory lift and G4 flux**: The F-theory interpretation of the su(10) F11 fibration (Section 5) admits G4-flux configurations satisfying all constraints simultaneously without a perturbative heterotic description. The Mordell-Weil rank of the elliptic fiber defines the U(1) gauge factors; GUT breaking to $\mathrm{SU}(5) \times \mathrm{U}(1)_Y$ follows from the $\mathrm{SU}(10) \supset \mathrm{SU}(5) \times \mathrm{U}(1)_Y$ decomposition along the fiber.

8.3 Why $h^{11} = 25$–28 is the productive band

The sweet spot $h^{1,1} = 25$–28 arises from the intersection of four independent constraints:

1. **Effective Picard number**: $h^{1,1}_{\mathrm{eff}} \approx 20$–$22$ is required for Yukawa rank in the sweet spot 140–159 (the rank of the $3h^{1,1}_{\mathrm{eff}} \times 3h^{1,1}_{\mathrm{eff}}$ Yukawa matrix). At $h^{1,1}_{\mathrm{eff}} < 18$ the Yukawa matrix is too small for a hierarchy; at $h^{1,1}_{\mathrm{eff}} > 22$ the compute wall is hit.
2. **LVS geometry**: LVS moduli stabilisation requires at least two del Pezzo divisors (as the instanton sources that generate the non-perturbative superpotential) and a K3-like divisor with $c_2 \cdot D \geq 24$ (setting the LVS instanton sector scale). Both conditions are generically satisfied at $h^{1,1} \geq 22$ and become increasingly rare below $h^{1,1} = 18$.

3. **Fibration richness**: The K3/elliptic fibrations required for SU(5) GUT gauge algebra (via the Kodaira I_5/I_{10} singularity) are present in $\sim 80\%$ of T3-verified entries at $h^{1,1} = 24$–28 but only $\sim 50\%$ at $h^{1,1} = 21$–23, and zero at $h^{1,1} \leq 16$ (cf. h16/P118).
4. **T0 gate**: Essentially no polytope at $h^{1,1} \geq 30$ satisfies $h^{1,1}_{\text{eff}} \leq 22$, eliminating the entire KS tail above the boundary.

8.4 F-theory interpretation

In the F-theory lift, the Calabi–Yau fourfold X_4 is an elliptic fibration over a base threefold B_3, with $h^{1,1}(X_4) = h^{1,1}(B_3) + 1$. The champion h26/P11670 with its su(10) fibration (Kodaira I_{10} singularity) is a candidate $\mathrm{SU}(10) \supset \mathrm{SU}(5) \times \mathrm{U}(1)_Y$ F-theory GUT over a base with $h^{1,1}(B_3) = 25$.

Tate's algorithm on the champion. The Weierstrass model of an elliptic fibration $\pi : X \to B$ is written as

$$y^2 = x^3 + f\,x\,z^4 + g\,z^6, \quad [x : y : z] \in \mathbb{P}^2_{1,2,3}, \tag{19}$$

where $f \in H^0(B, K_B^{-4})$ and $g \in H^0(B, K_B^{-6})$. The discriminant $\Delta = 4f^3 + 27g^2$ vanishes along the divisor containing the singular fiber. Tate's algorithm [12] reads off the Kodaira type from the vanishing orders $(\mathrm{ord}(a_1), \mathrm{ord}(a_2), \mathrm{ord}(a_3), \mathrm{ord}(a_4), \mathrm{ord}(a_6))$ of the Tate coefficients along the discriminant locus. For an I_{10} fiber: $\mathrm{ord}(\Delta) = 10$, $\mathrm{ord}(f) = 0$, $\mathrm{ord}(g) = 0$, yielding gauge algebra $\mathfrak{g} = \mathfrak{su}(10)$. The ambiguous F8 fiber (I_8 vs. III^*) is resolved by checking whether $\mathrm{ord}(a_6) \geq 2$: if yes, Kodaira type is III^* ($\mathfrak{e}_7$); if no, it is I_8 ($\mathfrak{su}(8)$) [12]. This is a finite computation on the toric data and is left for future work.

G4 flux and tadpole. In F-theory on X_4, the tadpole condition becomes

$$N_{\mathrm{D3}} + \frac{1}{2}\int_{X_4} G_4 \wedge G_4 = \frac{\chi(X_4)}{24}, \tag{20}$$

where $G_4 \in H^{2,2}(X_4; \mathbb{Z})$ is the four-form flux. The D3-brane tadpole obstruction found in the heterotic description (Section 7.3) translates to a constraint on G_4: for the GUT fibration, G_4 must be chosen to cancel the $\mathrm{ch}_2(V)$ contribution via the spectral cover construction. The spectral cover for SU(5) bundles in F-theory factorises as $\mathcal{C}_5 = 5\sigma + \eta$, where σ is the zero-section of π and $\eta \in H^2(B_3; \mathbb{Z})$ is the spectral class [13]. The heterotic tadpole constraint (14) in the F-theory language becomes $\eta = c_2(T_{B_3}) +$ corrections, which constrains the spectral class in terms of the base geometry. For h26/P11670, $c_2(T_{B_3})$ (with $h^{1,1}(B_3) = 25$) is computable from the toric fan of B_3, but the explicit identification of B_3 as a toric threefold is left for future work.

The Mordell–Weil rank bound from the chapter fibrations on P11670 is 4, providing enough U(1) factors for hypercharge assignment and proton-decay suppression [14].

The 780 entries with `has_GUT`= 1 in the T3 tier represent an unprecedented catalogue of F-theory GUT candidates in the $\chi = -6$ slice, all with verified triangulation stability and a quantified SM physics score. Of these, 37 are T4-verified with 200 FRST samples.

8.5 Open questions

Several physically important questions remain open:

1. **Full vacuum on h26/P11670**: Can the D3-tadpole obstruction be circumvented by the F-theory F11 fibration with an appropriate G4-flux choice? The I_{10} fiber admits a spectral cover construction that may provide additional tadpole contributions.

2. **Monad bundles at higher $k_{\max}$**: The scan at $k_{\max} = 2$ finds 612 slope-feasible candidates, all tadpole-obstructed. Raising to $k_{\max} = 3$ increases the number of slope-feasible candidates but also enlarges $\mathrm{ch}_2(V)$, making tadpole satisfaction harder. A rigorous $k_{\max} = 3$ scan is needed.
3. **Weierstrass resolution of F8 fibre**: The $\mathrm{I}_8/\mathrm{III}^*$ ambiguity on h26/P11670 resolves to either su(8) (matter representations $\mathbf{28} + \mathbf{8}$) or e_7 (matter $\mathbf{56}$). Tate's algorithm on the explicit toric variety settles this.
4. **Coverage closure at $h^{1,1} = 26$–$\mathbf{28}$**: h26 is 51% covered; the T0-wall analysis suggests the uninspected 49% is barren, but a definitive scan would confirm this.
5. **EFF_MAX extension**: Raising $h^{1,1}_{\mathrm{eff,max}}$ from 22 to 25 at $h^{1,1} = 30$–35 would test whether the landscape boundary is an artefact of the gate or a genuine physics phenomenon. Expected compute: $\sim 10\times$ per polytope.

9 Conclusion

We have performed the most extensive systematic survey to date of the $\chi = -6$ Kreuzer-Skarke landscape, processing 3.11M polytopes through a five-tier physics pipeline with explicit 100-point SM composite scoring.

Key results:

1. **Global optimum**: h26/P11670 (score 89), T4-verified over 200 triangulations, with SM and GUT fibration content including an su(10) GUT fiber (Section 5).
2. **Productive boundary**: $h^{1,1} \leq 28$; full scan of 5.45M polytopes at $h^{1,1} = 29$–32 returns zero entries above score 70.
3. **Low-$h^{1,1}$ champion**: h16/P118 (score 76) is the strongest SM candidate below $h^{1,1} = 20$, achieving its score without any fibration gauge structure — confirming the high-score landscape is genuinely concentrated at $h^{1,1} = 21$–28.
4. **T3 completeness**: all 938 score-$\geq$ 70 entries T3-verified; two entries improved at T3 (h23/P36: 79 $\to$ 81, h21/P9085: 79 $\to$ 80); T2 scoring is a reliable lower bound.
5. **Score stability**: T3$\to$T4 produces zero score changes; T3 depth (50 samples) is sufficient for ranking.
6. **Fibration richness**: 2,463 fibrations catalogued; 780 of 938 T3 entries carry SM+GUT gauge content (83%); 28 of 37 top-tier entries T4-verified with both SM and GUT fibers.
7. **D3-tadpole obstruction**: perturbative SU(4) monads on h26/P11670 are exactly obstructed; the F-theory lift via the su(10) F11 fibration is the most promising path to a full vacuum.

The pipeline, database, and KS index are released as open-source artefacts at https://github.com/sethc5/cytools_project.

A CYTools API Pitfalls

Nine non-obvious behaviours encountered during development:

Table 10: CYTools API pitfalls and workarounds.

#	Symptom	Root cause	Fix
A1	`fetch_polytopes()` returns ≤ 1000	Hidden default `limit=1000`	Pass
A2	FRST non-deterministic between runs	`get_cy()` uses random seed internally	Alwa fore
A3	c_2 wrong on non-favorable polytopes	Blow-down map not applied	Use
A4	Yukawa tensor not symmetric	Non-smooth FRSTs have wedge degeneracies	Filte metr
A5	Score drifts on `UPSERT`	sm_score recomputed from local T2 data on upsert	Post
A6	`get_hodge_numbers()` hangs	No timeout; CYTools CGI timeout on large polytopes	Sign
A7	$c_2 \cdot D < 0$ for some divisors	Expected: rigid divisors have $c_2 \cdot D < 0$	Docu
A8	Fibration id-collision on `INSERT`	Old merge script retained raw DB `id`	Strip poly
A9	`--local-ks` polytope ordering differs from server	CYTools server applies its own sort	All same

The remainder of this appendix details each pitfall with a minimal reproducing example and the resolution adopted.

A1. `fetch_polytopes()` hidden limit

CYTools' `fetch_polytopes(h11=N)` silently caps the return at 1000 polytopes by default. The symptom is that scanning $h^{1,1} = 22$ with no explicit limit returns exactly 1000 entries regardless of the KS-file size, and the DB count never exceeds 1000 for any $h^{1,1}$. The fix is straightforward:

```
# Wrong:
polys = cy.fetch_polytopes(h11=22)
# Correct:
polys = cy.fetch_polytopes(h11=22, limit=300000)
```

In our pipeline we always pass `limit=ks_index.get_count(h11)` which reads the pre-built index to obtain the exact polytope count.

A2. FRST non-determinism

`Polytope.get_cy()` calls an internal random triangulation routine whose output depends on the Python/NumPy random state. Two pipeline runs on the same polytope can return different FRSTs (and thus different Hodge numbers on non-favorable polytopes). This caused score drift of ± 2–4 between checkpointed runs. The fix is to always call `np.random.seed(k)` immediately before each `get_cy()` call with a deterministic seed derived from the polytope index:

```
import numpy as np
np.random.seed(int(poly_idx) % (2**31))
cy = poly.get_cy()
```

T3 and T4 tiers deliberately *vary* the seed (seeds $0, \ldots, n-1$) to sample the triangulation distribution.

A3. c_2 wrong on non-favorable polytopes

On polytopes where $h^{1,1}(X) < h^{1,1}(V_\Delta)$ (non-favorable embedding), the divisor classes of X form a proper sublattice of the toric ambient-space divisors. Calling `cy.second_chern_class()` without first applying the blow-down map returns c_2 in the ambient basis, giving wrong intersection numbers $c_2 \cdot D_i$. The correct call is:

```
divs = cy.get_cy_divisors()   # returns basis in X lattice
c2   = cy.second_chern_class()
c2_vals = [c2.dot(d) for d in divs]
```

This affected $\sim 3\%$ of T1 entries in early runs; all were recomputed after the fix.

A4. Yukawa tensor asymmetry

For FRSTs with non-simplicial cones, the holomorphic $(3,0)$-form Ω_3 does not localize cleanly on each 3-cone, causing the three-point integral $Y_{abc} = \int \Omega_3 \wedge \psi_a \wedge \psi_b \wedge \psi_c$ to be non-symmetric under permutation of a, b, c. In practice we observed $|Y_{abc} - Y_{bac}|/|Y_{abc}| \leq 0.1$ for smooth FRSTs and up to 0.4 for non-smooth ones. The resolution is to: (i) prefer smooth FRSTs (`cy.is_smooth()` check); (ii) symmetrize $Y \leftarrow (Y + Y^T)/2$ post-hoc for the Yukawa-hierarchy score; and (iii) record both the raw asymmetry and the symmetrized hierarchy in the DB.

A5. Score drift on `UPSERT`

Early versions of the pipeline recomputed `sm_score` from the local T2 data during every `db.upsert_polytope()` call, using the formula stored at the time of the upsert. When the score formula was updated (e.g. the `yukawa_rank` sweet-spot tier widened from 120–149 to 140–159 in patch v5.2), previously upserted entries retained the old score until the next upsert, creating two generations of score values in the DB simultaneously. The fix is a post-upsert rescore pass: run `tools/rescore_all.py` to uniformly reapply the current formula to all DB entries before any analysis. All reported scores in this paper reflect a single rescore pass at formula version v5.2.

A6. `get_hodge_numbers()` hang

CYTools' `Polytope.get_hodge_numbers()` calls an external CGI service with no client-side timeout. On large polytopes ($h^{1,1} \geq 30$) this call occasionally hangs indefinitely, blocking the pipeline worker. We wrap the call in a Python signal-based timeout:

```
import signal
def _timeout(signum, frame): raise TimeoutError
signal.signal(signal.SIGALRM, _timeout)
signal.alarm(30)            # 30-second hard limit
try:
    h = poly.get_hodge_numbers()
finally:
    signal.alarm(0)
```

Timed-out polytopes are logged and skipped; they represent $< 0.01\%$ of the scanned range.

A7. $c_2 \cdot D < 0$ for rigid divisors

For divisors D that contract to a point (rigid divisors, $h^0(\mathcal{O}_D) = 1$), the intersection $c_2 \cdot D$ is expected to be *negative* by adjunction: $c_2 \cdot D = 2\chi(D) - D^3$ with $\chi(D) = 1$ for a rigid surface and $D^3 < 0$. Early pipeline versions logged these as errors and skipped the polytope. The fix is to *document* this as expected behaviour and record separate columns `n_rigid`, `c2_D_min` in the DB schema. Rigid divisors are excluded from the LVS instanton count (they do not contribute an ADS superpotential term) but remain useful as obstructions to fibration structure.

A8. Fibration id-collision on `INSERT`

The initial DB merge script preserved the raw SQLite `id` primary-key column from the source DB. On merge, duplicate fibration rows (same polytope, same fiber type) with different `ids` caused a primary-key collision, and the merge failed silently (SQLite `INSERT OR IGNORE`). This left some polytopes with zero fibration entries despite having valid fibers in the source shard. The fix is to strip the `id` column from shards before merge and deduplicate on the semantic key `(h11, poly_idx, fiber_type, fiber_surface)`:

```
INSERT OR IGNORE INTO fibrations
  (h11, poly_idx, fiber_type, fiber_surface,
   kodaira_types, gauge_algebra, ...)
SELECT h11, poly_idx, fiber_type, fiber_surface,
       kodaira_types, gauge_algebra, ...
FROM   shard_fibrations;
```

A9. `--local-ks` polytope ordering

CYTools' KS server applies an internal sort (ascending Minkowski-sum volume) before returning polytopes. Our local KS static files are sorted by KS index, which *correlates* with but does not *equal* the server sort. Running the pipeline once with the server and once with `--local-ks` produced polytopes in different order, so a "resume from checkpoint" run skipped previously scanned indices when the two sources were mixed. The fix: all pipeline runs within a project must use the *same* KS source (either server throughout or local throughout). All results in this paper use `--local-ks` exclusively, with the local KS index built from the 2024-06-01 static snapshot of the Kreuzer–Skarke database.

B Full T4 Results Table

All 37 score-$\geq$ 80 entries T4-verified (200 triangulations, 60-sample stability recompute, commit `d97d158`, 2026-03-06). Zero score changes from T3 to T4.

Table 11: T4-verified entries (score $\geq$ 80). Columns: $h^{1,1}$, poly_idx, score, SM, GUT, n_{clean}, c_2-stable, κ-stable.

$h^{1,1}$	poly_idx	score	SM	GUT	n_{cl}	c_2-stab	κ-stab
26	11670	**89**	✓	✓	22	0.033	0.033
29	8423	87	✓	✓	30	0.000	0.000
24	868	87	✓	✓	24	0.017	0.017
27	9192	87	✓	✓	22	0.000	0.000
27	4102	87	–	–	22	0.000	0.000
22	682	85	✓	✓	84	0.000	0.000
28	6642	85	✓	✓	30	0.000	0.000
27	9133	85	✓	✓	22	0.000	0.000
24	45873	85	✓	✓	22	0.000	0.000
25	46481	85	✓	✓	22	0.000	0.000
27	45013	85	✓	✓	12	0.000	0.000
29	13253	83	✓	✓	20	0.000	0.000
28	5473	82	✓	✓	26	0.000	0.000
27	2317	82	✓	✓	24	0.000	0.000
28	718	82	–	–	22	0.000	0.000
28	33	82	–	–	20	0.017	0.017

$h^{1,1}$	poly_idx	score	SM	GUT	n_{cl}	c_2-stab	κ-stab
23	36	81	–	–	60	0.000	0.000
25	860	81	✓	✓	24	0.100	0.133
25	7867	81	✓	✓	18	0.000	0.000
27	27537	81	✓	✓	16	0.000	0.000
27	39352	81	✓	✓	12	0.000	0.000
21	9085	80	–	–	66	0.017	0.017
26	315	80	✓	✓	32	0.000	0.000
28	33562	80	✓	✓	28	0.000	0.000
25	5449	80	✓	✓	26	0.000	0.000
24	44004	80	✓	✓	26	0.000	0.000
26	11871	80	✓	✓	26	0.000	0.000
25	38242	80	✓	✓	24	0.000	0.000
25	8995	80	✓	✓	24	0.000	0.000
24	272	80	–	–	24	0.000	0.017
29	6577	80	✓	✓	22	0.083	0.083
29	6575	80	✓	✓	22	0.083	0.083
25	18950	80	✓	✓	22	0.000	0.000
26	30513	80	✓	✓	22	0.000	0.000
28	1937	80	✓	✓	20	0.000	0.000
27	28704	80	✓	✓	20	0.000	0.000
27	26021	80	✓	✓	20	0.000	0.000

C Database Schema

```
CREATE TABLE polytopes (
  h11 INTEGER, poly_idx INTEGER,
  h21 INTEGER, chi INTEGER,
  h11_eff INTEGER, gap INTEGER,
  n_clean INTEGER, n_bundles_checked INTEGER,
  max_h0 INTEGER, first_clean_at INTEGER,
  n_dp INTEGER, n_k3_div INTEGER, n_rigid INTEGER,
  has_swiss INTEGER, n_swiss INTEGER,
  best_swiss_tau REAL, volume_hierarchy REAL,
  yukawa_texture_rank INTEGER, yukawa_hierarchy REAL,
  yukawa_zeros INTEGER,
  n_mori_rays INTEGER, n_dp_contract INTEGER,
  n_k3_fib INTEGER, n_ell_fib INTEGER, n_fibers INTEGER,
  has_SM INTEGER, has_GUT INTEGER, best_gauge TEXT,
  d3_min REAL, d3_max REAL, d3_n_distinct INTEGER,
  sm_score INTEGER, n_triangulations INTEGER,
  tri_c2_stable_frac REAL, tri_kappa_stable_frac REAL,
  tier_reached TEXT, last_updated TEXT,
  UNIQUE(h11, poly_idx)
);

CREATE TABLE fibrations (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  h11 INTEGER, poly_idx INTEGER,
  fiber_type TEXT, fiber_surface TEXT,
```

```
  kodaira_types TEXT,   -- JSON: [{type, algebra, rank, confidence}]
  gauge_algebra TEXT, gauge_rank INTEGER,
  contains_SM INTEGER, has_SU5_GUT INTEGER,
  MW_rank_bound INTEGER
);
```

D SM Score Component Derivations

This appendix derives the calibration of each of the ten score components (Table 2), motivating the thresholds and weights from physical requirements.

D1. Yukawa hierarchy (30 pts)

The observed top-quark Yukawa coupling $y_t \approx 1$ with bottom-quark $y_b \approx 0.02$ sets a hierarchy of order 50. Including tau-lepton and neutrino Yukawa structure, a viable three-generation model needs $\Delta_Y = \lambda_{\max}/\lambda_{\min}$ of the 3×3 Yukawa matrix to satisfy $\Delta_Y \gtrsim 10^2$. The score is assigned logarithmically:

$$s_Y = 30 \cdot \min\left(1,\ \frac{\log_{10} \Delta_Y}{\log_{10} 2000}\right),$$

so that $\Delta_Y = 2000$ achieves the maximum 30 pts, $\Delta_Y = 100$ gives ≈ 19 pts, and $\Delta_Y = 1$ (flat matrix) gives 0 pts. The threshold 2000 was calibrated by surveying the distribution of Δ_Y across the T2-scored dataset: the 95th percentile is $\Delta_Y \approx 1800$, making 2000 a mild stretch goal.

D2. Yukawa rank (15 pts)

The Yukawa rank component measures the total Frobenius norm $\|Y\|_F = \sqrt{\sum_{a,b} Y_{ab}^2}$. A sweet spot of 140–159 is identified empirically: below 140 the Yukawa couplings are too weak to produce viable masses; above 159 the matrix is too degenerate (all singular values comparable, no hierarchy). The score is:

$$s_R = \begin{cases} 15 & 140 \leq \|Y\|_F \leq 159 \\ 10 & 100 \leq \|Y\|_F < 140 \text{ or } 160 \leq \|Y\|_F < 200 \\ 5 & \|Y\|_F < 100 \text{ or } \|Y\|_F \geq 200 \\ 0 & \text{no Yukawa computed} \end{cases}$$

D3. LVS quality (15 pts)

The LVS quality component encodes the Swiss-cheese viability score computed from the Kähler potential (3). It is graded into five tiers by the number of identified Swiss-cheese configurations n_{Swiss} and the best small-cycle ratio $\tau_s^*/\mathcal{V}^{2/3}$:

n_{Swiss}	$\tau_s^*/\mathcal{V}^{2/3}$	Score
≥ 20	> 0.01	15
10–19	> 0.01	12
5–9	> 0.01	8
1–4	any	4
0	—	0

The threshold $n_{\text{Swiss}} \geq 20$ is motivated by the requirement that the LVS minimum be accessible from multiple Kähler cone chambers, making the result robust to FRST variation.

D4. Clean bundles (10 pts)

Each clean bundle (with $h^0 = 3$ and all higher cohomology zero) represents a candidate three-generation sector. The score increases with n_{clean}:

$$s_{\text{cl}} = \begin{cases} 10 & n_{\text{clean}} \geq 60 \\ 8 & 22 \leq n_{\text{clean}} < 60 \\ 5 & 10 \leq n_{\text{clean}} < 22 \\ 2 & 1 \leq n_{\text{clean}} < 10 \\ 0 & n_{\text{clean}} = 0 \end{cases}$$

The thresholds 22 and 60 correspond to the 75th and 95th percentiles of the n_{clean} distribution across T2-scored entries.

D5–D10. Remaining components

D5. Volume hierarchy (7 pts): the ratio $\mathcal{V}_b/\mathcal{V}_s$ measured in the balanced Kähler cone, graded by thresholds $\{10^3, 10^4, 10^5, 10^6, 10^7\}$ (1 pt per threshold crossed). The seven-point scale mirrors the LVS requirement of $\mathcal{V} \sim e^{a\tau_s}$ with $a\tau_s \sim 7$–14 for $m_{3/2} \in [1\,\text{TeV}, 10^{13}\,\text{GeV}]$.

D6. D3 diversity (5 pts): the count of distinct D3-brane charges $\{Q_{\text{D3}}(L_\alpha)\}$ among clean bundles. A larger diversity set enables more distinct tadpole-cancellation solutions, each corresponding to an independent flux sector. Score: 1 pt per distinct charge, capped at 5 pts (5 charges).

D7. Clean depth (5 pts): the census index of the *first* clean bundle found, normalised so that index ≤ 10 (within the first 10 line bundles scanned) gives 5 pts. Finding clean bundles early suggests a dense distribution, reducing dependence on the arbitrary census bound $K_{\max}$.

D8. Clean rate (5 pts): the fraction $n_{\text{clean}}/n_{\text{checked}}$ of the census, graded at thresholds $\{2\%, 5\%, 10\%, 20\%, 30\%\}$. High clean rate indicates a geometrically favourable divisor lattice.

D9–D10. Bundle quality (8 pts combined): composite metrics combining D7 and D8 with a rank-sweet-spot bonus (3 pts for $\|Y\|_F \in [140, 159]$, counted separately as D9). The 8-point combined bundle quality rewards polytopes with both high rate and early discovery.

Score formula stability. The formula was frozen at version v5.2 (2026-02-28) and applied uniformly to all DB entries via a single rescore pass. No retroactive component reweighting was performed after the first T3-tier scan; all reported scores are v5.2 values. The champion score of 89 is stable: a $\pm 10\%$ perturbation of each component weight shifts the h26/P11670 rank by at most 1 position among the top-10 entries.

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